

# TenderHub Deployment Manual

Production Cloud Architecture, Deployment Blueprint & Operations Guide

OFFICIAL SPECIFICATION

|                                                                                       |                                                                       |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>APPLICATION ARCHITECTURE</b><br>Monorepo: Backend API + Frontend SPA + Admin Panel | <b>PRIMARY CLOUD PLATFORM</b><br>Render / Railway / Cloudflare Pages  |
| <b>DATABASE &amp; CACHE</b><br>MongoDB Atlas (M0/M10) + Upstash Redis                 | <b>STORAGE LAYER</b><br>Dual Cloudflare R2 Buckets (Primary & Backup) |

## 1. Production Topology & Service Boundaries

TenderHub is split into three decoupled components to ensure high availability, fast asset delivery, and operational security:

| Service Name     | Directory            | Type                                  | Production Host                       | Internal Port       |
|------------------|----------------------|---------------------------------------|---------------------------------------|---------------------|
| tenderhub-api    | backend/             | Node.js / Express Web Service         | Render / Railway / AWS EC2            | Port 8000           |
| tenderhub-web    | frontend/            | React 19 / Vite Static Client         | Cloudflare Pages / Vercel / Render    | Port 5173 / CDN     |
| tenderhub-admin  | admin-panel/         | React 19 / Vite Admin Dashboard       | Private Subdomain / Cloudflare Access | Port 5174 / CDN     |
| tenderhub-worker | backend/src/workers/ | BullMQ Asynchronous Background Worker | Background Worker / Daemon Process    | Headless Playwright |

## 2. Option A: 1-Click Deployment via Render Blueprint (render.yaml)

The repository root contains an automated deployment specification ( `render.yaml` ).

**Deployment Steps:**

- Push repository to GitHub: `git push origin main`
- Open [Render Dashboard](#) > Click **New +** > Select **Blueprint**.
- Connect repository: Render automatically detects `render.yaml` .
- Provide required secrets ( `MONGO_URI` , `REDIS_URL` , `R2_*` ).
- Click **Apply**. Render builds and launches both the Web Service and Static Site concurrently.

## 3. Terminal Commands for Generating Both PDF Manuals

**Terminal Automation:** You can regenerate either or both instruction manuals anytime directly from the terminal inside the `backend` directory.

# 1. Generate Disaster Recovery & Backup Manual (PDF): `npm run manual:disaster` # 2. Generate Production Deployment & Operations Manual (PDF): `npm run manual:deployment` # 3. Generate BOTH Official Manuals Simultaneously: `npm run manuals:all`

# TenderHub Deployment Manual

Environment Variables, Security Hardening & Health Telemetry

CONFIGURATION & AUDIT

## 4. Production Environment Variables Reference

| Variable Name        | Service Scope    | Production Value / Recommendation                         |
|----------------------|------------------|-----------------------------------------------------------|
| PORT                 | Backend          | 8000 (or auto-assigned by cloud provider)                 |
| NODE_ENV             | Backend          | production                                                |
| MONGO_URI            | Backend          | MongoDB Atlas Primary Connection String                   |
| SECONDARY_MONGO_URI  | Backend          | Automated Fallback Cluster (Instant Failover)             |
| REDIS_URL            | Backend / Worker | Upstash Serverless Redis or Redis Cloud URI               |
| R2_ACCOUNT_ID        | Backend          | Primary Cloudflare Account ID ( tenderhub )               |
| BACKUP_R2_ACCOUNT_ID | Backend          | Secondary Cloudflare Account ID ( tenderhub-backup )      |
| ADMIN_SECRET_KEY     | Backend / Admin  | 32-character random string for Admin Panel auth           |
| VITE_API_URL         | Frontend / Admin | Public Backend HTTPS URL (e.g. https://api.tenderhub.in ) |

## 5. Post-Deployment Verification & Health Check Checklist

- Health Endpoint:** Test https://your-backend.com/health (returns 200 OK with uptime).
  - Database Connectivity:** Check logs for MongoDB connected successfully [PRIMARY DB] .
  - Redis & BullMQ Queues:** Check logs for Redis Connected Successfully .
  - Scheduler Confirmation:** Verify all 5 automated scraping slots (09:00, 10:00, 13:00, 15:00, 18:30) and 02:00 AM retention cron are active.
  - Verify Admin Panel Telemetry:** Log in to http://localhost:5174 or your admin subdomain, open Telemetry & Health and Backup & Recovery.

**Engineering Support:** Both manual PDFs ( TenderHub\_Disaster\_Recovery\_and\_Backup\_Manual.pdf and TenderHub\_Production\_Deployment\_Manual.pdf ) are archived in the docs/ folder of the repository.